YEHIA HOSSAM

Data Engineer

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Introduction

High-impact Data Engineer with over 3 years of experience architecting scalable data infrastructure, enterprise ETL pipelines, and warehouse solutions. Proven track record of optimizing large-scale environments at Teradata for critical national projects and managing end-to-end data life-cycles using SQL, Python, and Apache NIFI. Adept at cross-functional collaboration to serve clean, optimized data layers for Software, BI, and AI/ML engineering teams.

Experience

Data Engineer 12/2025 – Present
Teradata Cairo, Egypt

- Administrative Control Authority (ACA) Engineered enterprise data integration workflows to ingest and consolidate 40+ governmental data sources across Egypt into a unified, large-scale Teradata Data Warehouse.
- Optimized existing ETL processes and SQL queries, rewriting legacy transformation logic to boost pipeline efficiency and data ingestion speed by 25%.
- Designed and maintained high-performance Stored Procedures (SPs) for software engineering teams, validating backend API integrations and ensuring seamless application performance.
- Owned the end-to-end deployment lifecycle, managing the testing and validation of data models in the Dev environment before coordinating and executing heavy production loads.
- Diagnosed and resolved production bugs routed from the Managed Services team, collaborating with the Data Modeling team to implement structural SQL fixes and prevent data pipeline regressions.
- Acted as the primary technical bridge for BI and AI teams; unblocked BI engineers by solving immediate reporting layer bottlenecks, and engineered custom data sources to feed downstream NLP and data-matching AI models.

Data Analytics Engineer 05/2023 – 11/2025
Marketeers Research Cairo, Egypt

- Designed and implemented the enterprise star schema data warehouse for audit sales data from scratch utilizing MySQL and AWS RDS, establishing the company's foundational analytics infrastructure.
- Scaled and enhanced the data warehouse architecture to seamlessly integrate complex new data streams, including e-commerce platforms and retail APIs using Python.
- Refactored and optimized legacy ETL pipelines and automation scripts, slashing data delivery times by 50% and significantly increasing pipeline reliability across reporting layers.
- Built robust Python-based automation modules and integrated the Google Forms API with Excel templates to automate reporting from SSI software and standardize data workflows.
- Applied strict clean code practices and database indexing tuning techniques to improve long-term code-base maintainability and boost overall query speed.

Education

Computer and Systems Engineering, Zagazig University 08/2017 – 07/2022

- **Grade: 81%**
- **Graduation Project Grade:** A+

Projects

Bike Sales Data Mart.

- Design Date Warehouse(DWH) Star schema for sales and making ETL Using SSIS
- **Tools Used:** Sql Server - DWH - Data Mart - ETL - SSIS - Data Modeling - Star schema - Fact table- Dimension tables - Dashboard - Power Bi.

Nyc-taxi Data Flow And Analysis

- Designing and developing a whole data flow starting from generating data from API then building orchestration using Mage AI to automate generating, sending output to Google storage and from google storage to google big-query then making data model using DBT and finally using Google looker studio to visualize data.
- **Tools Used:** Docker - Mage AI - GCP - Google big-query - Python - SQL - dbt - Google looker studio - google sheets.

E-commerce DWH

- Using Kimball approach to design DWH to serve data that comes from DWH website making model and querying to answer business questions.
- **Tools Used:** MySQL - Python - pandas - Data Modeling - Star schema

Server metrics streaming with kafka

- Receiving stream from 10 clusters simulated by java app add them to SQL database and process the data using pyspark and save result in Hadoop HDFS.
- **Tools Used:** Kafka - pyspark - big data - json - MySQL - docker - hadoop.

Skills

Programming Languages: Python - R - Dax - SQL.
Data Engineering: ETL - Teradata SQL- DWH - MongoDb - SSIS - DBT - SSMS - Google big-query - RBDMS - Airflow - NIFI.
Big Data: Hadoop - Spark - Kafka
Data Analysis: Excel - SQL - Python - PowerBI - Tableau - Looker studio.
Data Science: Math - Statistics - Machine learning - Numpy -Pandas - Seaborn - Matplotlib - Sklearn.
Tools: Git - Visual Code - Docker - Visual Studio - SSMS -AWS - GCP.
Concept: OOP - Algorithms - Problem solving - Linux.

Certificates

- Fabric Data Engineer Associate DP-700 03/2026
- Microsoft Azure Data Fundamentals DP-900 06/2025
- Apache Airflow 3 Fundamentals 06/2025
- AWS Cloud Practitioner 03/2025
- ALX Data Science 08/2024
- HackerRank SQL(Advanced) 01/2023

Extra-Curricular Activities

- IEEE RAS - Electrical Team Lead 02/2019 - 06/2020
- AIESEC - Team lead 01/2019 - 12/2019

Languages

- Arabic : Native
- English : B2